

Omar Benali

Senior AI Solutions Architect | Boston, MA | (617) 445-2891 | omar.benali@emailprovider.com

PROFESSIONAL EXPERIENCE

Senior AI Solutions Architect | BioTech Innovations Labs | Boston, MA | January 2024 – Present

- Large language models, generative AI frameworks
- RAG systems, vector databases, semantic search
- Healthcare data pipelines, HIPAA compliance architecture
- Claude API, GPT-4, LangChain integration points
- Prompt engineering, few-shot learning methodologies
- Biomedical NLP applications, clinical note processing
- Azure OpenAI deployment configurations

AI Research Specialist | LifeScience Analytics Group | Cambridge, MA | July 2022 – December 2023

- Machine learning model evaluation, ensemble techniques
- TensorFlow, PyTorch implementation frameworks
- Data annotation workflows, medical imaging datasets
- Hugging Face transformers, fine-tuning approaches
- Genomic data preprocessing, feature extraction
- REST API design patterns, microservices architecture
- Docker containerization, Kubernetes orchestration concepts

Machine Learning Developer | Wellness Data Systems | Providence, RI | March 2021 – June 2022

- Python, scikit-learn, pandas, NumPy libraries
- SQL database design, query optimization
- Patient outcome prediction models
- Data visualization tools: Tableau, Matplotlib, Seaborn
- A/B testing frameworks, statistical analysis
- AWS S3, EC2, RDS services overview
- Git version control, CI/CD pipeline fundamentals

Junior Data Analyst | Regional Health Diagnostics | Worcester, MA | September 2019 – February 2021

- Excel, Python scripting for data processing
- Clinical trial data management, validation protocols
- SQL queries, database maintenance
- Report generation, stakeholder presentations
- CSV file handling, data normalization
- Basic statistical analysis, descriptive metrics
- Laboratory information systems (LIS)

TECHNICAL SKILLS

AI & Machine Learning: GPT-4, Claude API, LangChain, LlamaIndex, Anthropic SDK, Hugging Face Transformers, BERT, LoRA fine-tuning, retrieval-augmented generation, prompt engineering, few-shot learning, chain-of-thought reasoning

Deep Learning Frameworks: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost

Data & Databases: PostgreSQL, MongoDB, Elasticsearch, Pinecone, Weaviate, DuckDB, Apache Spark

Cloud & DevOps: AWS (S3, EC2, Lambda, SageMaker), Azure OpenAI, Google Cloud Platform, Docker, Kubernetes

Languages: Python, SQL, JavaScript, Bash

Healthcare & Life Sciences: HIPAA compliance, clinical NLP, biomedical data standards, genomic analysis, EHR integration, medical imaging, HL7 protocols

Tools & Platforms: Jupyter Notebook, VS Code, Weights & Biases, MLflow, DVC, Git, Figma, Notion

EDUCATION

Master of Science, Data Science | Northeastern University | Boston, MA | 2019

Bachelor of Science, Biology | University of Massachusetts Amherst | Amherst, MA | 2017